Reducing Cardiovascular Disease in Indonesia

Health Impact, Costs, and Cost-Effectiveness of Clinical and Public-Health Interventions, 2025–2050

Disease Control Priorities

Briefing for the Ministry of Health, Republic of Indonesia

August 2026

The challenge, and the opportunity

Most premature cardiovascular deaths in Indonesia can be prevented or treated

Cardiovascular disease is Indonesia's largest health burden

Under current trends, a person reaching age 30 faces roughly a **14.6% chance** of dying from cardiovascular disease before age 70. Most of this is **preventable or treatable**.

- Premature CVD deaths (ages 30–69) have risen for two decades and remain high.
- Two complementary levers:
  - **Clinical care** — prevent and better treat established disease.
  - **Population prevention** — tobacco, salt, trans-fat.
- This analysis weighs their **health impact and value for money** together.

Source: *Indonesia CVD model, calibrated to GBD 2023 and UN World Population Prospects 2024; combined results workbook (annual probability-of-death sheet).* ***Method:*** *the probability of death from CVD before age 70 is the chance of dying from the six modelled cardiovascular diseases between exact ages 30 and 70, read from a single-year life table in the 2025 baseline year.*

We measured the health gains and the value for money of proven interventions

General aim

Estimate the **health impact and value for money** of scaling up evidence-based **clinical** and **public-health** cardiovascular interventions in Indonesia through **2050**.

Aim 1 — Burden

Project the CVD burden and the probability of death from CVD before age 70 under the **current course** and under **alternative intervention scenarios**.

Aim 2 — Value for money

Estimate **deaths averted**, mortality reductions, **additional costs**, and the **cost per death averted** for each intervention and package.

Aim 3 — Better together

Quantify the **added value of combining** clinical and public-health action, modelled as a single joint run rather than a sum.

Source: *study design; Indonesia CVD model.* ***Method:*** *every scenario runs through one projection over 2025–2050; the baseline (current course) is the no-new-intervention comparator against which each scenario is measured.*

Together, clinical care and prevention could avert about 5.0 million deaths by 2050

Combining clinical + public health is the big prize

Together they could avert about **5.0M deaths** over 2025–2050, cut the probability of death from CVD before age 70 in 2050 by **32%**, at about **US$8,279 per death averted**.

- **All clinical interventions**: $\approx$3.2M deaths averted; probability of death from CVD before age 70 down 23% by 2050.
- **All public-health policies**: $\approx$2.1M deaths averted at about US$156 per death averted.
- **Most impactful single levers**: Primary CVD prevention and Tobacco tax.
- **Affordability**: the combined package costs about US$8,279 per death averted.

Source: *results contract (`09_deck_results.rds`) + combined workbook.* ***Method:*** *deaths averted cumulative 2025–2050 (six CVD causes + type-2 diabetes); cost per death averted = discounted additional cost ÷ deaths averted (2023 US$).* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Without discounting, combining care and prevention still averts about 5.0 million deaths by 2050

Combining clinical + public health, valued undiscounted

Together they could avert about **5.0M deaths** over 2025–2050, cut the probability of death from CVD before age 70 in 2050 by **32%**, at about **US$13,486 per death averted (undiscounted)**.

- **All clinical interventions**: ≈3.2M deaths averted; probability of death from CVD before age 70 down 23% by 2050.
- **All public-health policies**: ≈2.1M deaths averted at about US$229 per death averted (undiscounted).
- **Most impactful single levers**: Primary CVD prevention and Tobacco tax.
- **Affordability (undiscounted)**: the combined package costs about US$13,486 per death averted.

Source: *results contract (`09_deck_results.rds`) + combined workbook.* ***Method:*** *deaths averted cumulative 2025–2050 (six CVD causes + type-2 diabetes);* ***undiscounted*** *cost per death averted =* ***undiscounted*** *additional cost ÷ deaths averted (2023 US$, health system). Health results are identical to the discounted slide.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

The model follows how prevention and treatment change disease and survival

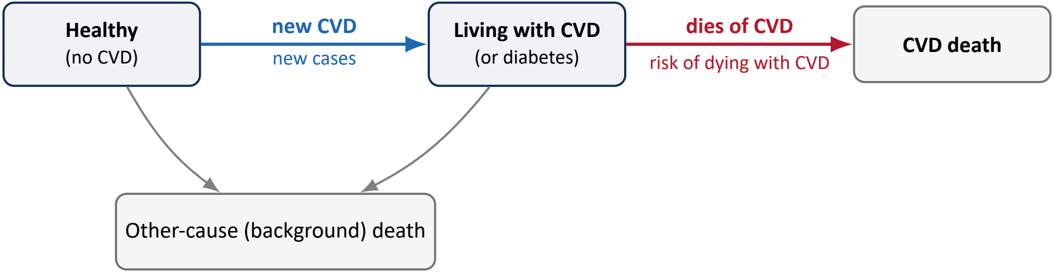
1. **People & demography.** Indonesia's population is projected by age and sex (UN World Population Prospects 2024).
2. **Starting health.** Disease and death rates are set to Indonesia's data (GBD 2023) and calibrated to match observed levels.
3. **Year-by-year simulation.** Each year people can develop cardiovascular disease or diabetes, survive, or die.
4. **Interventions.** Clinical and public-health scenarios change how many people develop disease (new cases) and how many survive it.
5. **Outcomes.** The model reports deaths, healthy years lost or gained (DALYs), the probability of death from CVD before age 70, costs, and value for money.

One consistent engine

Every scenario runs through the **same** projection, so results are directly comparable and the **combined** package is modelled as one run — not a sum of separate estimates.

Source: *Indonesia CVD model engine; GBD 2023 (starting rates); UN World Population Prospects 2024 (demography, life expectancy).* ***Method:*** *a discrete-time state-transition simulation by single-year age and sex, calibrated to match observed GBD levels using a standard numerical optimiser.*

Interventions work by preventing new disease and by helping people survive it



Fewer people fall ill
Primary prevention & *all* public-health policies reduce **new cases**.

More people survive
Secondary prevention, treatment & tobacco policy lower the **risk of dying after developing CVD**.

Background mortality
Other-cause deaths from demography; unchanged by these interventions.

Source: *Indonesia CVD model engine (`06_run_scenarios_indonesia_fair.R`).* ***Method:*** *interventions act on two transitions — new cases (Healthy → Living with CVD) and the risk of dying after developing CVD (Living with CVD → death); no recovery/remission is modelled, a conservative assumption.*

The interventions we tested

Clinical care prevents disease and helps people survive it

Intervention	Target population	Model effect
Primary CVD prevention	Adults at cardiovascular risk	Fewer new cases
Type-2 diabetes treatment	People with type-2 diabetes	Better survival
Treatment of acute coronary syndromes	Target patient group	Better survival
Heart-attack secondary prevention	Heart-attack (IHD) survivors	Better survival
Acute heart-failure care	Acute heart-failure patients	Better survival
Chronic heart-failure care	Chronic heart-failure patients	Better survival
Stroke secondary prevention	Stroke survivors	Better survival

Implementation scenario (all clinical interventions): coverage scales up **from a low base toward 80%** of the people who need each service between 2025 and 2050. Primary prevention lowers who develops CVD (new cases); secondary prevention and treatment improve survival.

Source: *clinical results workbook, sheets `Selected_Interventions` and `Scenario_Catalog`.* ***Method:*** *the table lists the clinical interventions active in this run (excluded interventions never appear); effect sizes come from the FAIR Choices intervention taxonomy.*

Population policies cut tobacco, salt, and trans-fat risk across the whole population

Policy	Risk factor	Model effect	Implementation lever
Tobacco tax	Tobacco	New cases + survival	Raise tax to 75% of retail price
Tobacco policy package	Tobacco	New cases + survival	Clean-air law, mass media, ad ban

Policies reduce exposure (smoking, salt, trans-fat intake), lowering the number of **new CVD cases**. **Tobacco** policies additionally improve **survival** after CVD. Packages are modelled as single joint runs of their component measures.

Source: *public-health results workbook, sheets `Selected_Interventions` / `Policy_Levers` and `Scenario_Catalog`.*
Method: *the table lists the policies active in this run; effect sizes come from published risk-factor evidence stored in the workbook.*

Every input is traceable to a recognised data source

Model component	Primary source	Years / version	Use in model
Population & demography	UN World Population Prospects	2024 revision	Population, background mortality, life expectancy
Cause-specific mortality & disease rates	GBD (Indonesia)	2023	Baseline burden & calibration targets
Disability weights (healthy years lost)	GBD	2023	Healthy years lost or gained (DALYs)
Clinical intervention effects	FAIR Choices intervention taxonomy	current	New-case & survival effects
Public-health effect sizes	Published risk-factor evidence (in workbook)	current	New-case & survival effects
Intervention & policy costs	Programme & policy unit costs (in workbook)	price year 2023	Budget impact & value for money
International CVD mortality comparison	WPP-scaled ASEAN comparison series	1995–2050	Benchmarking Indonesia vs. neighbours

Population: UN World Population Prospects 2024; epidemiology: GBD 2023 (GBD 2023 Cardiovascular Diseases and Risks Collaborators 2025; Schumacher et al. 2024).

Source: *modelling pipeline and input workbooks.* ***Method:*** *these inputs drive the demographic projection, calibration targets, and effect sizes; effect-size and cost citations are in the workbooks.*

Costs depend on who needs care, how many receive it, and local prices

We estimate the **additional health-system resources required to expand effective coverage**, using a costing approach adapted from Disease Control Priorities (DCP) (Watkins et al. 2020).

People who need the service → **Current coverage** & scale-up target → **Services** per person / year → **Indonesia unit** cost → **Annual cost** baseline & scenario → **Additional & dis-counted cost;** cost per person; cost per death prevented

What this tells policymakers: the analysis estimates how much additional health-system spending is required to move from today's coverage toward universal, high-quality access—and relates that investment to the health gains achieved.

Source: *intervention-based costing adapted from Watkins et al. (2020); unit costs and coverage from the input workbooks; demography from UN WPP 2024.* ***Method:*** *annual cost = people in need × coverage × annual service frequency × Indonesia-adjusted unit cost; additional cost = scenario minus baseline; equations in the appendix.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

What the interventions deliver, and at what cost

Scaling up clinical care sharply cuts premature cardiovascular deaths

Clinical intervention	Deaths averted (cumulative)	Prob. of CVD death before age 70, 2050	Reduction in that probability, 2050	Deaths reduction 2050	Cost per death averted	Avg. annual add'l cost/capita, 2026-50 (disc.)	Add'l cost/capita in 2050 (disc.)
Primary CVD prevention	2,134,000	9.6%	18.1%	12.5%	US$8,608	US$2.52	US$3.52
Type-2 diabetes treatment	458,000	11.6%	0.5%	2.4%	US$15,314	US$0.96	US$1.46
Treatment of acute coronary syndromes	315,000	11.3%	3.1%	1.4%	US$23,164	US$1.00	US$1.50
Heart-attack secondary prevention	158,000	11.5%	1.6%	0.7%	US$32,038	US$0.70	US$1.07
Acute heart-failure care	99,000	11.6%	0.6%	0.5%	US$30,783	US$0.42	US$0.66
Chronic heart-failure care	67,000	11.6%	0.6%	0.3%	US$20,607	US$0.19	US$0.29
Stroke secondary prevention	37,000	11.7%	0.3%	0.1%	US$109,233	US$0.55	US$0.78
All clinical (combined)	**3,176,000**	**9.0%**	**23.4%**	**17.1%**	**US$13,014**	**US$5.67**	**US$7.98**

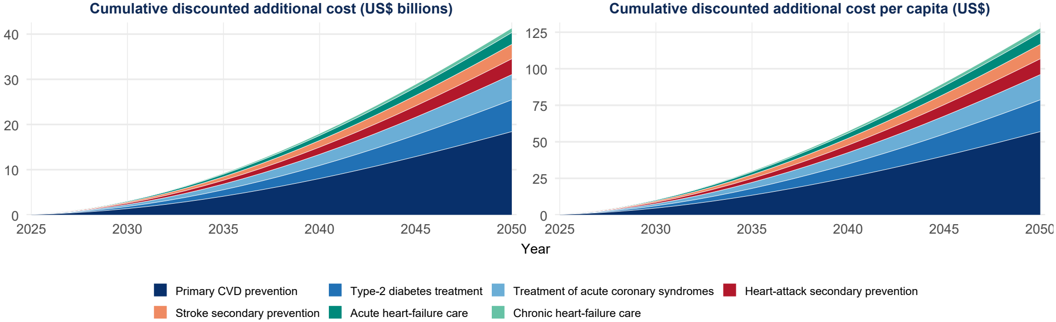
Source: *clinical cost workbook (`indonesia_model_cost_value_formulae.xlsx`, sheet `Budget_Impact`); population is the workbook's own `national_population`.* ***Method:*** *cost per death averted = discounted additional cost ÷ deaths averted (3%/yr, 2023 US$, health system).* ***Avg. annual additional cost per capita, 2026–2050*** *= cumulative discounted additional cost over 2026–2050 ÷ (25 × 2026 population).* ***Additional cost per capita in 2050*** *= discounted additional cost in 2050 ÷ 2050 population.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Undiscounted costs are higher, yet clinical care still sharply cuts premature deaths

Clinical intervention	Deaths averted (cumulative)	Prob. of CVD death before age 70, 2050	Reduction in that probability, 2050	Deaths reduction 2050	Cost per death averted (undisc.)	Avg. annual add'l cost/capita, 2026-50 (undisc.)	Add'l cost/capita in 2050 (undisc.)
Primary CVD prevention	2,134,000	9.6%	18.1%	12.5%	US$14,021	US$4.11	US$7.38
Type-2 diabetes treatment	458,000	11.6%	0.5%	2.4%	US$25,216	US$1.59	US$3.05
Treatment of acute coronary syndromes	315,000	11.3%	3.1%	1.4%	US$38,092	US$1.65	US$3.14
Heart-attack secondary prevention	158,000	11.5%	1.6%	0.7%	US$52,818	US$1.15	US$2.23
Acute heart-failure care	99,000	11.6%	0.6%	0.5%	US$50,904	US$0.69	US$1.38
Chronic heart-failure care	67,000	11.6%	0.6%	0.3%	US$34,061	US$0.31	US$0.62
Stroke secondary prevention	37,000	11.7%	0.3%	0.1%	US$178,379	US$0.90	US$1.64
All clinical (combined)	**3,176,000**	**9.0%**	**23.4%**	**17.1%**	**US$21,213**	**US$9.26**	**US$16.70**

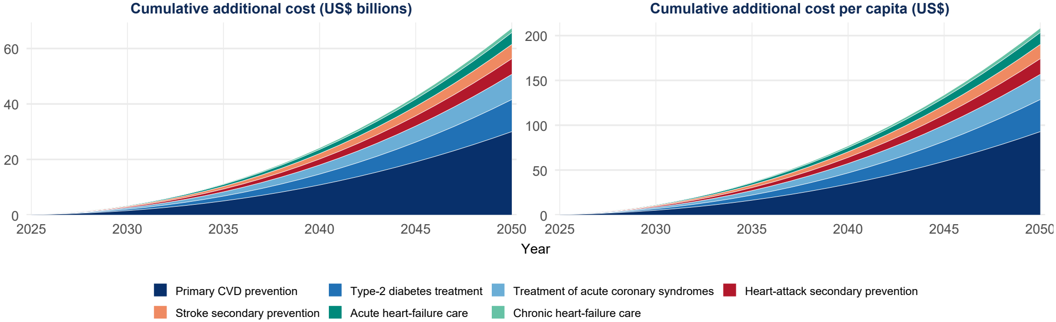
Source: *clinical cost workbook (`indonesia_model_cost_value_formulae.xlsx`, sheet `Budget_Impact`); population is the workbook's own `national_population`.* ***Method (undiscounted):*** *cost per death averted =* ***undiscounted*** *additional cost ÷ deaths averted (2023 US$, health system).* ***Avg. annual additional cost per capita, 2026–2050*** *= cumulative* ***undiscounted*** *additional cost over 2026–2050 ÷ (25 × 2026 population).* ***Additional cost per capita in 2050*** *=* ***undiscounted*** *additional cost in 2050 ÷ 2050 population. Health results are identical to the discounted slide.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Primary CVD prevention drives most of Indonesia's clinical cardiovascular budget through 2050



Source: *clinical cost workbook (`indonesia_model_cost_value_formulae.xlsx`, sheet `Annual_Cost`); all-clinical scenario, discounted additional cost by intervention (shared components counted once); population is the workbook's own `national_population`.* ***Method:*** *2025 is the discount base year and coverage scales up over 2025–2050; left is cumulative discounted additional cost through each year, right is that total ÷ that year's population; intervention totals reconcile to the all-clinical totals by year.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Primary CVD prevention still drives most of the undiscounted clinical cardiovascular budget through 2050



Source: *clinical cost workbook (`indonesia_model_cost_value_formulae.xlsx`, sheet `Annual_Cost`); all-clinical scenario,* ***undiscounted*** *additional cost by intervention (shared components counted once); population is the workbook's own `national_population`.* ***Method (undiscounted):*** *coverage scales up over 2025–2050; left is cumulative* ***undiscounted*** *additional cost through each year, right is that total ÷ that year's population (2023 US$); intervention totals reconcile to the all-clinical undiscounted totals by year.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Population policies avert deaths at very low cost

Public-health policy	Deaths averted (cumulative)	Prob. of CVD death before age 70, 2050	Reduction in that probability, 2050	Deaths reduction 2050	Cost per death averted	Avg. annual add’l cost/capita, 2026-50 (disc.)	Add’l cost/capita in 2050 (disc.)
Tobacco tax	1,453,000	10.8%	7.9%	5.0%	US$141	US$0.028	US$0.019
Tobacco policy package	650,000	11.3%	3.6%	2.2%	US$182	US$0.016	US$0.011
All public health (combined)	**2,071,000**	**10.4%**	**11.2%**	**7.1%**	**US$156**	**US$0.045**	**US$0.030**

Source: *public-health cost workbook (`indonesia_cost_value_public_health_formulae.xlsx`); annual additional cost reproduced from its `Annual_Cost` inputs and reconciled to the run total; population is the workbook’s own `national_population`.* ***Method:*** *cumulative deaths averted 2025–2050 vs baseline; tobacco and sodium packages each modelled as one joint run.* ***Avg. annual additional cost per capita, 2026–2050*** *= cumulative discounted additional cost over 2026–2050 ÷ (25 × 2026 population);* ***cost per capita in 2050*** *= discounted additional cost in 2050 ÷ 2050 population.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Even undiscounted, population policies avert deaths at very low cost

Public-health policy	Deaths averted (cumulative)	Prob. of CVD death before age 70, 2050	Reduction in that probability, 2050	Deaths reduction 2050	Cost per death averted (undisc.)	Avg. annual add'l cost/capita, 2026-50 (undisc.)	Add'l cost/capita in 2050 (undisc.)
Tobacco tax	1,453,000	10.8%	7.9%	5.0%	US$207	US$0.041	US$0.040
Tobacco policy package	650,000	11.3%	3.6%	2.2%	US$267	US$0.024	US$0.023
All public health (combined)	**2,071,000**	**10.4%**	**11.2%**	**7.1%**	**US$229**	**US$0.065**	**US$0.064**

Source: *public-health cost workbook (`indonesia_cost_value_public_health_formulae.xlsx`); annual additional cost reproduced from its `Annual_Cost` inputs and reconciled to the run's **undiscounted** total; population is the workbook's own `national_population`. **Method (undiscounted):** cumulative deaths averted 2025–2050 vs baseline; tobacco and sodium packages each modelled as one joint run. **Avg. annual additional cost per capita, 2026–2050** = cumulative **undiscounted** additional cost over 2026–2050 ÷ (25 × 2026 population); **cost per capita in 2050** = **undiscounted** additional cost in 2050 ÷ 2050 population (2023 US$). Health results are identical to the discounted slide. **Preliminary costing results:** unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Clinical and population action, combined

Clinical care and prevention achieve more together than alone

Package	Deaths averted (cumulative)	Reduction in prob. of CVD death <70, 2050	Prob. of CVD death before age 70, 2050	Discounted total cost	Cost per death averted	Avg. annual add'l cost/capita, 2026-50 (disc.)	Add'l cost/capita in 2050 (disc.)
All clinical (combined)	3,176,000	23.4%	9.0%	US$41.3B	US$13,014	US$5.67	US$7.98
All public health (combined)	2,071,000	11.2%	10.4%	US$323.6M	US$156	US$0.045	US$0.030
Clinical + public health (combined)	**4,996,000**	**31.8%**	**8.0%**	**US$41.4B**	**US$8,279**	**US$5.67**	**US$7.97**

The combined package is modelled as one run — not a sum

Because the two families' effects overlap, the joint scenario is projected in a single run: about **5.0M deaths** averted and the 2050 probability of CVD death before age 70 cut to **8.0%** (from 11.7%), at about **US$8,279 per death averted**.

Source: *combined cost workbook (`indonesia_model_cost_value_clinical_public_health_formulae.xlsx`); annual additional cost reproduced from its `Annual_Cost` inputs and reconciled to the run total; population is the workbook's own `national_population`.* ***Method:*** *joint package = one projection applying both families (not their sum).* ***Avg. annual additional cost per capita, 2026–2050*** *= cumulative discounted additional cost over 2026–2050 ÷ (25 × 2026 population);* ***cost per capita in 2050*** *= discounted additional cost in 2050 ÷ 2050 population (3%/yr, 2023 US$).* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Undiscounted, clinical care and prevention still achieve more together

Package	Deaths averted (cumulative)	Reduction in prob. of CVD death <70, 2050	Prob. of CVD death before age 70, 2050	Undiscounted total cost	Cost per death averted (undisc.)	Avg. annual add'l cost/capita, 2026-50 (undisc.)	Add'l cost/capita in 2050 (undisc.)
All clinical (combined)	3,176,000	23.4%	9.0%	US$67.4B	US$21,213	US$9.26	US$16.70
All public health (combined)	2,071,000	11.2%	10.4%	US$475.1M	US$229	US$0.065	US$0.064
Clinical + public health (combined)	**4,996,000**	**31.8%**	**8.0%**	**US$67.4B**	**US$13,486**	**US$9.26**	**US$16.69**

The combined package is one run — valued undiscounted

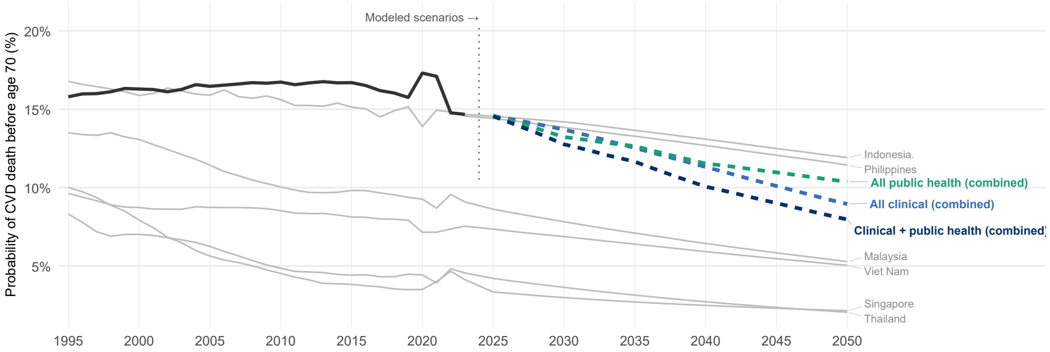
Because the two families' effects overlap, the joint scenario is projected in a single run: about **5.0M deaths** averted and the 2050 probability of CVD death before age 70 cut to **8.0%** (from 11.7%), at about **US$13,486 per death averted (undiscounted)**.

Source: *combined cost workbook (`indonesia_model_cost_value_clinical_public_health_formulae.xlsx`); annual additional cost reproduced from its `Annual_Cost` inputs and reconciled to the run's* ***undiscounted*** *total; population is the workbook's own `national_population`.* ***Method (undiscounted):*** *joint package = one projection applying both families (not their sum).* ***Avg. annual additional cost per capita, 2026–2050*** *= cumulative* ***undiscounted*** *additional cost over 2026–2050 ÷ (25 × 2026 population);* ***cost per capita in 2050*** *=* ***undiscounted*** *additional cost in 2050 ÷ 2050 population (2023 US$). Health results are identical to the discounted slide.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Indonesia’s premature cardiovascular mortality can fall toward its neighbours

Probability of death from CVD before age 70: observed, projected, and neighbours

Chance of dying from CVD between exact ages 30 and 70



Indonesia observed history (solid, to 2023) and ASEAN comparators are WPP-scaled reference series.
Modelled intervention scenarios are dashed for 2025–2050.
ASEAN comparator trends use linear extrapolation for illustrative reference.

Source: *combined results workbook (annual probability-of-death sheet, R-reconciled values); ASEAN comparator series in* `data/raw/`. ***Method:*** *model scenarios (dashed) run 2025–2050 for the six modelled cardiovascular causes over exact ages 30–70; comparators are WPP-scaled reference series shown for context only.*

A stronger care cascade: the 70-30-30 to 70-70-70 strategy

The cascade scale-up is ambitious but staged

Hypertension and type-2 diabetes are modelled with **diagnosed** $\rightarrow$ **treated** $\rightarrow$ **controlled** care cascades; **cholesterol** treatment coverage follows hypertension coverage. Coverage rises in two stages, then holds.

Two staged milestones

- **2030 — 70-30-30:** 70% diagnosed, 30% of those treated, 30% of those controlled.
- **2040 — 70-70-70:** 70% diagnosed, 70% treated, 70% controlled; held through 2050.
- Linear scale-up from the observed baseline; coverage already above a milestone is **not scaled down**.

Shares of all people with the condition

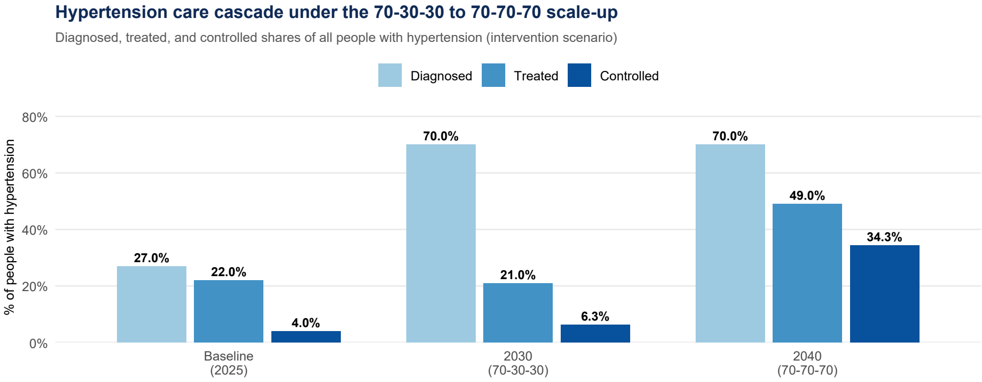
	2030	**2040**
Diagnosed	70%	70%
Treated	21%	49%
Controlled	6.3%	34.3%

treated $= 70\% \times \{30, 70\}\%$; controlled $=$ treated $\times\{30, 70\}\%$.

Effective modelled coverage $=$ controlled $+\frac{1}{2}\times$ treated-but-uncontrolled $=$ **13.65%** (2030) and **41.65%** (2040): full effect to controlled patients, half to treated-but-uncontrolled.

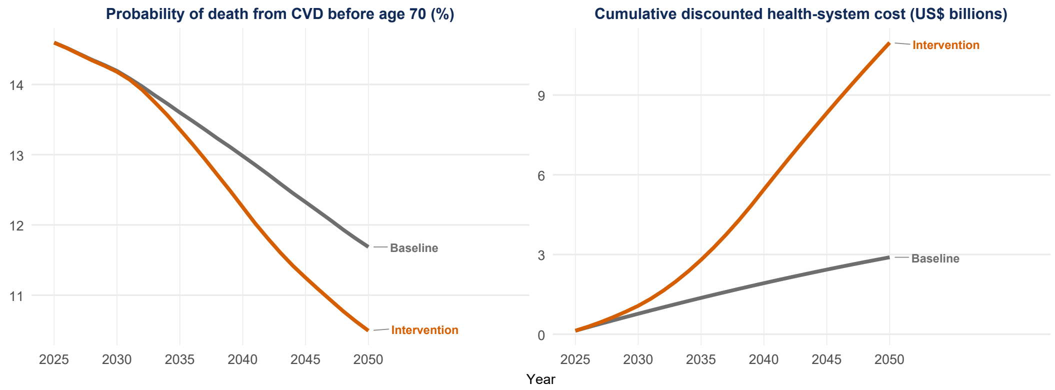
Source: *cascade workbook (`Cascade_Assumptions`, `Cascade_Trajectory`); isolated runner; NCD-RisC coverage inputs.*
Method: *nested targets on the diagnosed/treated/controlled shares; piecewise-linear scale-up, no backsliding.*

Indonesia’s hypertension cascade can progress in two achievable stages



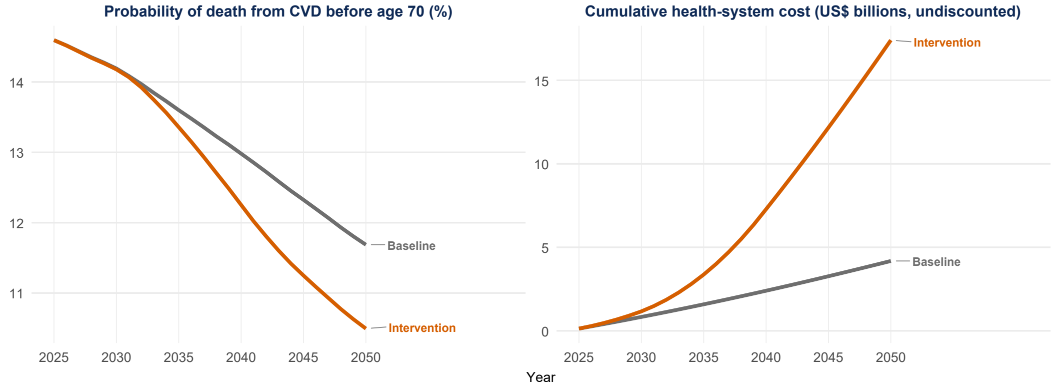
Source: *cascade `Cascade_Trajectory` (sex-averaged) and `Cascade_Assumptions`; scale-up begins 2026.* ***Method:*** *baseline (2025) shows only measured treated/controlled — diagnosis is not modelled at baseline, so no diagnosed bar. Type-2 diabetes follows the same targets (baseline treated ≈ 5.6%, controlled ≈ 1.2%).*

Stronger prevention lowers premature deaths at a rising but bounded cost



Source: *cascade workbook — annual probability-of-death sheet (R-reconciled) and `Annual_Cost`, cumulatively summed and reconciled to `Budget_Impact`.* ***Method:*** *probability over the six CVD causes, ages 30–70; cumulative discounted health-system cost (3%/yr to 2025, 2023 US$) of the CVD primary-care and type-2 diabetes cascades.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Undiscounted, stronger prevention still lowers premature deaths at a rising cost



Source: *cascade workbook — annual probability-of-death sheet (R-reconciled) and `Annual_Cost`, cumulatively summed and reconciled to `Budget_Impact` **undiscounted** cumulative additional cost.* ***Method:*** *probability over the six CVD causes, ages 30–70 (identical to the discounted slide); cumulative **undiscounted** health-system cost (2023 US$) of the CVD primary-care and type-2 diabetes cascades.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

A stronger cascade could prevent about 1.5 million deaths by 2050

Cumulative from 2025 through year (unless noted)	**2030**	**2040**	**2050**
Projected deaths — CVD (six causes) + type-2 diabetes (million)			
Baseline (current course)	6.106	18.396	33.345
Intervention	6.092	18.014	31.832
Projected health-system cost (US$ billion, discounted)			
Baseline (current course)	0.77	1.92	2.89
Intervention	1.07	5.45	10.97
Net effect of the strategy			
Additional cost (US$ billion, discounted)	0.30	3.53	8.08
Deaths prevented (million)	0.014	0.382	1.513
Annual financing intensity (that year only, not cumulative)			
Additional cost per capita in that year (US$, discounted)	US$0.33	US$1.56	US$1.31

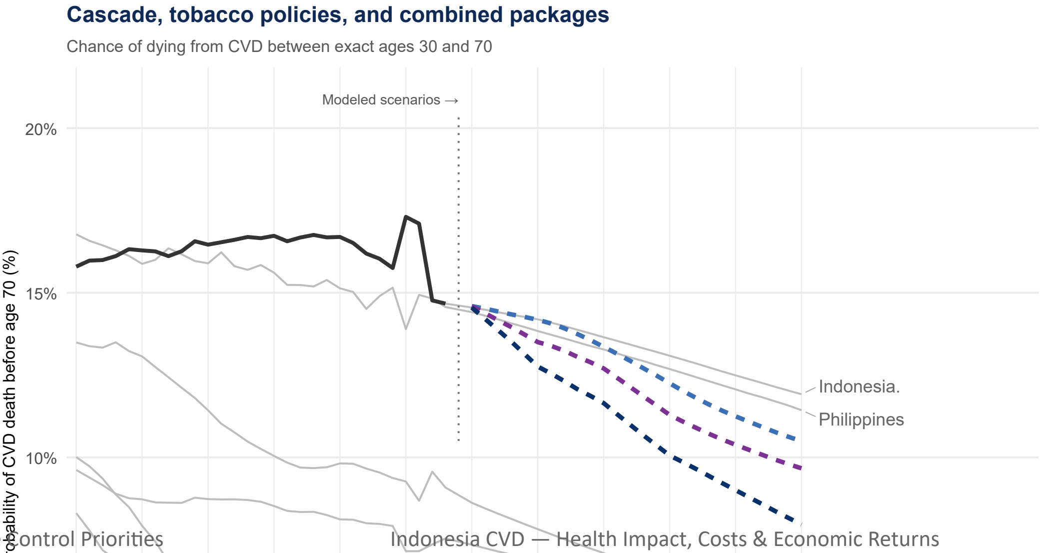
Source: *cascade `Annual_Mortality`, `Annual_Cost/Budget_Impact`.* ***Method:*** *deaths/cost columns cumulative from 2025 through that year (six CVD causes + type-2 diabetes; discounted 3% to 2025, 2023 US$); the per-capita row is that year only, not cumulative.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

On an undiscounted basis, a stronger cascade still prevents about 1.5 million deaths by 2050

Cumulative from 2025 through year (unless noted)	2030	2040	2050
Projected deaths — CVD (six causes) + type-2 diabetes (million)			
Baseline (current course)	6.106	18.396	33.345
Intervention	6.092	18.014	31.832
Projected health-system cost (US$ billion, undiscounted)			
Baseline (current course)	0.83	2.40	4.18
Intervention	1.16	7.28	17.40
Net effect of the strategy			
Additional cost (US$ billion, undiscounted)	0.33	4.88	13.22
Deaths prevented (million)	0.014	0.382	1.513
Annual financing intensity (that year only, not cumulative)			
Additional cost per capita in that year (US$, undiscounted)	US$0.39	US$2.43	US$2.74

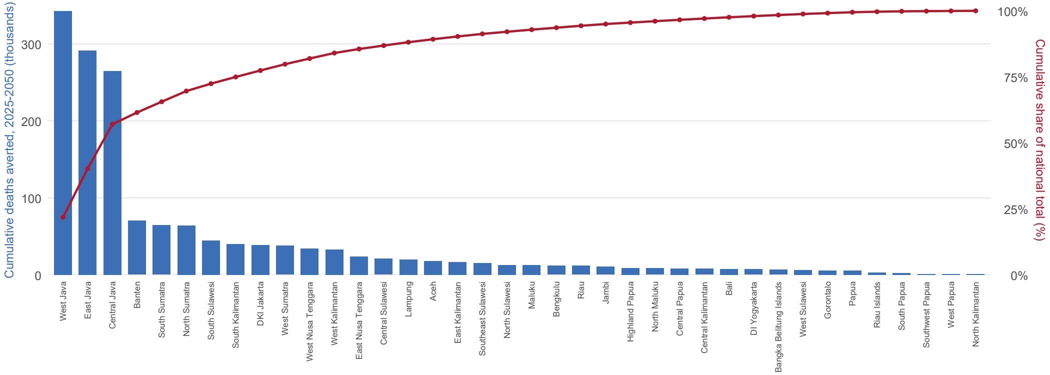
Source: *cascade `Annual_Mortality`, `Annual_Cost/Budget_Impact`.* ***Method (undiscounted):*** *deaths/cost columns cumulative from 2025 through that year (six CVD causes + type-2 diabetes;* ***undiscounted****, 2023 US$); the per-capita row is that year only, not cumulative. Deaths are identical to the discounted slide.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

The care cascade and tobacco policies jointly move Indonesia further toward its neighbours



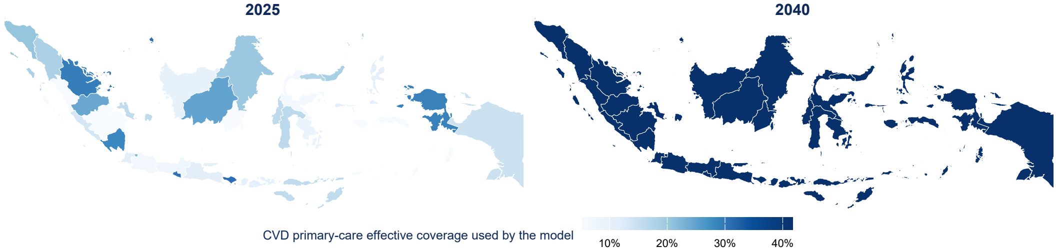
Where the gains fall: a province-level view of the cascade

About half of averted deaths come from just 3 of Indonesia's 38 provinces



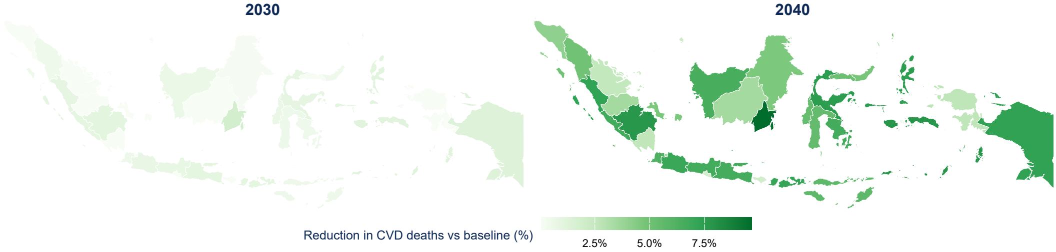
Source: *isolated subnational cascade workbook (`..._formulae_subnational.xlsx`), sheet `Health_Outcomes`.* ***Method:*** *bars are cumulative deaths averted 2025–2050 per province (official cascade scope: six CVD causes + type-2 diabetes); the line is the running share of the national total. Province-summed baseline deaths reconcile to `Province_Reconciliation` (PASS; residual -0.14%).* ***Preliminary—validate before use:*** *provincial baseline cascade estimates and province-adjusted costs require validation by Indonesia's Ministry of Health and BPJS. Provincial CVD treatment coverage is a crude anchor from observed CVD mortality and the modelled coverage-effect relationship (extreme values capped); provincial diabetes coverage is scaled in proportion.*

CVD primary-care coverage rises toward a common target in every province by 2040



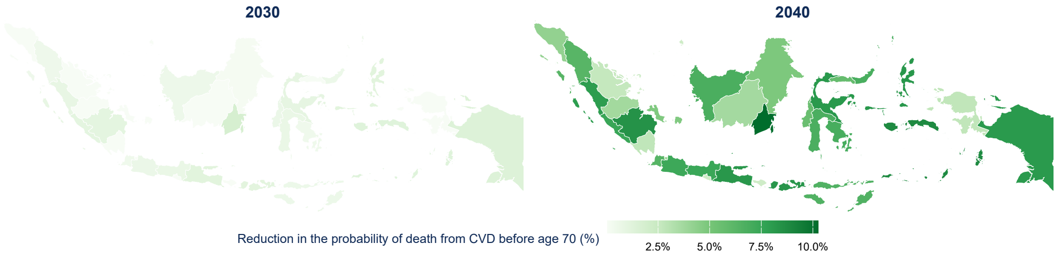
Source: *subnational cascade workbook, sheet `Cascade_Trajectory`.* ***Method:*** *CVD primary-care effective coverage (the model's own effect/cost driver) for 2025 and 2040; the workbook provides no sex-population split, so the two sexes are pooled with equal weight (their modelled coverage is identical from the 2040 target onward); one common scale. Boundaries: offline admin-1 polygons; the 5 post-2012 provinces appear within their parent region (an exact geographic subset).* ***Preliminary—validate before use:*** *provincial baseline cascade estimates and province-adjusted costs require validation by Indonesia's Ministry of Health and BPJS. Provincial CVD treatment coverage is a crude anchor from observed CVD mortality and the modelled coverage-effect relationship (extreme values capped); provincial diabetes coverage is scaled in proportion.*

Cardiovascular deaths fall most where the baseline gaps and burden are largest



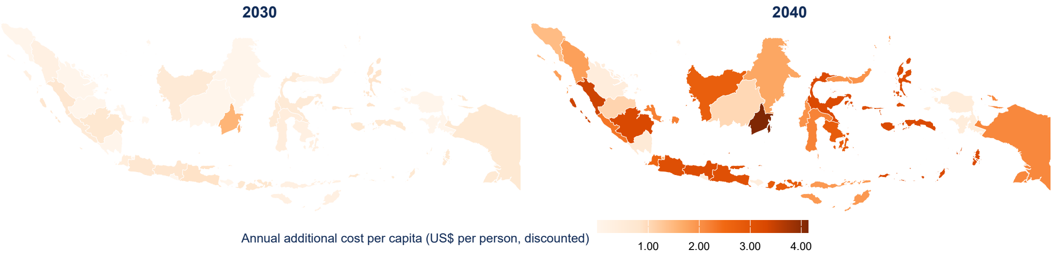
Source: *subnational cascade workbook, sheet* `Annual_Mortality`*.* ***Method:*** *percentage reduction in deaths from the six modelled CVD causes (type-2 diabetes excluded) = (baseline − scenario) ÷ baseline, for 2030 and 2040; one common scale. Boundaries: offline admin-1 polygons; the 5 post-2012 provinces appear within their parent region (an exact geographic subset).* ***Preliminary—validate before use:*** *provincial baseline cascade estimates and province-adjusted costs require validation by Indonesia's Ministry of Health and BPJS. Provincial CVD treatment coverage is a crude anchor from observed CVD mortality and the modelled coverage-effect relationship (extreme values capped); provincial diabetes coverage is scaled in proportion.*

Premature cardiovascular mortality falls in every province by 2040



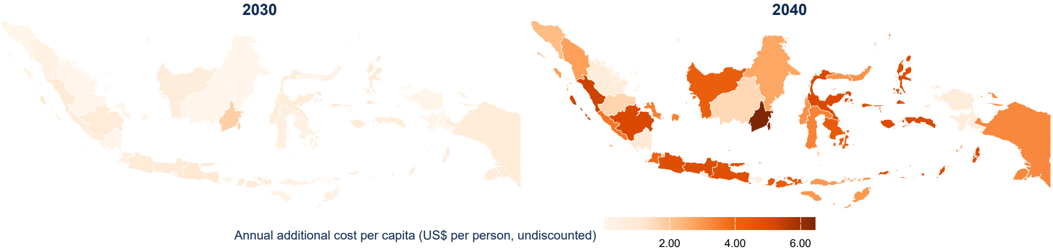
Source: *subnational cascade workbook, probability-of-death-before-70 sheet (R-reconciled literal probabilities).* ***Method:*** *percentage reduction in the probability of death from CVD before age 70 = (baseline − scenario) ÷ baseline, for 2030 and 2040, pooled to the boundary polygon with population weights; one common scale. Boundaries: offline admin-1 polygons; the 5 post-2012 provinces appear within their parent region (an exact geographic subset).* ***Preliminary—validate before use:*** *provincial baseline cascade estimates and province-adjusted costs require validation by Indonesia's Ministry of Health and BPJS. Provincial CVD treatment coverage is a crude anchor from observed CVD mortality and the modelled coverage-effect relationship (extreme values capped); provincial diabetes coverage is scaled in proportion.*

The provincial financing requirement grows as coverage scales up



Source: *subnational cascade workbook, sheets `Annual_Cost` / `Budget_Impact`.* ***Method:*** *annual (not cumulative) discounted additional cost in each province divided by that province's population, for 2030 and 2040 (discounted 3%/yr to 2025, 2023 US$); one common scale. Boundaries: offline admin-1 polygons; the 5 post-2012 provinces appear within their parent region (an exact geographic subset).* ***Preliminary—validate before use:*** *provincial baseline cascade estimates and province-adjusted costs require validation by Indonesia's Ministry of Health and BPJS. Provincial CVD treatment coverage is a crude anchor from observed CVD mortality and the modelled coverage-effect relationship (extreme values capped); provincial diabetes coverage is scaled in proportion.*

The undiscounted provincial financing requirement grows as coverage scales up



Source: *subnational cascade workbook, sheets `Annual_Cost` / `Budget_Impact`.* ***Method (undiscounted):*** *annual (not cumulative)* ***undiscounted*** *additional cost in each province divided by that province's population, for 2030 and 2040 (2023 US$); one common scale. Boundaries: offline admin-1 polygons; the 5 post-2012 provinces appear within their parent region (an exact geographic subset).* ***Preliminary—validate before use:*** *provincial baseline cascade estimates and province-adjusted costs require validation by Indonesia's Ministry of Health and BPJS. Provincial CVD treatment coverage is a crude anchor from observed CVD mortality and the modelled coverage-effect relationship (extreme values capped); provincial diabetes coverage is scaled in proportion.*

What this means for policy

Prevention and clinical care together offer the biggest, most affordable gains

- **Biggest health gains come from doing both.** The combined package averts far more (~5.0M) than either family alone.
- **Public-health policies are exceptional value.** Population-wide fiscal/regulatory measures avert deaths at a fraction of clinical cost per death averted.
- **Clinical scale-up delivers large, durable gains** in survival and premature mortality.
- **Sequence for early wins.** Tobacco taxation and other prevention policies can start cutting risk quickly and cheaply, while clinical capacity scales up.
- **The premature-mortality target is within reach.** The combined package bends Indonesia's probability of death from CVD before age 70 down by about 32% by 2050.

Source: *results contract (`09_deck_results.rds`); cascade workbook.* ***Method:*** *interpretation of modelled deaths averted, premature-mortality reduction, and cost per death averted; policy translation needs judgement on feasibility, financing, and delivery.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

What these results can and cannot tell us

- Results depend on **modelled epidemiological estimates** (GBD) and calibration; the real burden may differ.
- **Effect sizes, future coverage, and unit costs are uncertain**; interventions are assumed to reach and sustain their targets.
- The disease model is **simplified** (healthy $\rightarrow$ living with CVD $\rightarrow$ death; no recovery; no comorbidity interactions).
- Value for money is expressed as **cost per death averted**; effects on illness (morbidity) and productivity are not valued in money here.
- **Costing results are preliminary.** Unit costs, service use, and delivery assumptions require further validation against **BPJS and Ministry of Health** data before policy use.
- Projections are **scenarios, not forecasts or guarantees**; they show what is achievable if interventions are delivered as modelled.

Source: *Indonesia CVD model and its input workbooks (GBD 2023; UN WPP 2024; FAIR Choices effects; programme and policy unit costs).* ***Method:*** *qualitative statement of the assumptions and uncertainties that bound the quantitative results above.*

The case for action: two levers, affordable, and a reachable target

1. Both, together

Clinical + public health averts **~5.0M deaths** by 2050 — more than either alone.

2. Prevention is affordable

Population policies avert deaths at about **US$156** each.

3. A reachable target

The probability of death from CVD before age 70 falls **32%** by 2050 under the combined package.

Combined action against CVD could avert about **5.0M deaths** at about **US$8,279 per death averted** through 2050.

Source: *results contract (`09_deck_results.rds`).* ***Method:*** *deaths averted cumulative 2025–2050; reduction in the probability of death from CVD before age 70 in 2050; cost per death averted uses discounted additional cost.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Undiscounted, the case still holds: two levers, affordable, and a reachable target

1. Both, together

Clinical + public health averts **~5.0M deaths** by 2050 — more than either alone.

2. Prevention is affordable

Population policies avert deaths at about **US$229** each (undiscounted).

3. A reachable target

The probability of death from CVD before age 70 falls **32%** by 2050 under the combined package.

Combined action against CVD could avert about **5.0M deaths** at about **US$13,486 per death averted (undiscounted)** through 2050.

Source: *results contract (`09_deck_results.rds`).* ***Method (undiscounted):*** *deaths averted cumulative 2025–2050; reduction in the probability of death from CVD before age 70 in 2050; cost per death averted uses* ***undiscounted*** *additional cost (2023 US$). Health results are identical to the discounted slide.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Technical appendix

Appendix 1 — Scenario crosswalk

Family	Workbook scenario id	Executive label
Clinical (FAIR Choices)	`all`	All selected interventions (combined)
	`I_CVD_PRIMARY`	Primary prevention of CVD
	`I_HF_ACUTE`	Acute heart-failure management
	`I_HF_CHRONIC`	Chronic heart-failure management
	`I_IHD_ACUTE`	Treatment of acute coronary syndromes
	`I_IHD_SECONDARY`	Secondary prevention of ischemic heart disease
	`I_STROKE_SECONDARY`	Secondary prevention of stroke
	`I_T2D_TREATMENT`	Longitudinal treatment of type 2 diabetes
	`baseline`	Baseline (no new intervention)
Public health	`all_public_health`	All selected public-health interventions (combined)
	`I_PH_TOBACCO_POLICY`	Tobacco policy package
	`I_PH_TOBACCO_TAX`	Tobacco tax
	`I_PH_TOB_AD_BAN`	Tobacco advertising ban
	`I_PH_TOB_CLEAN_AIR`	Smoke-free / clean-air law
	`I_PH_TOB_MEDIA`	Tobacco mass-media campaign
Combined	`all_clinical_public_health`	All clinical + public-health interventions (combined)

Source: *combined workbook `Scenario_Catalog` (binding active-scenario contract).* ***Method:*** *identity/labels/family read from the current run; excluded interventions never appear; packages are single joint runs of their children (omitted from main tables to avoid double-counting).*

Appendix 2 — Definitions, horizons and assumptions

Metrics

- **Deaths averted (cumulative)** $= \sum_{t=2025}^{2050}(\text{baseline deaths}_t - \text{scenario deaths}_t)$, across modelled CVD & diabetes causes.
- **% reduction in deaths, 2050** $= 100 \times (\text{baseline} - \text{scenario})/\text{baseline in 2050}$.
- **Probability of death from CVD before age 70** $=$ probability of dying of CVD between exact ages 30 and 70 (six CVD causes, both sexes, single-year life table).
- **% reduction in that probability, 2050**: analogous to deaths.

Costing & value for money

- Horizon 2025–2050; cost discount rate 3%/yr; price year 2023.
- Costs in **market US$**; cost perspective: health system.
- **Cost per death averted** $=$ cumulative **discounted** additional (incremental) cost $\div$ cumulative **undiscounted** deaths averted.
- A "Dominant" scenario averts deaths while lowering cost; "not defined" means no deaths averted.

Model. Discrete-time state-transition model; single-year ages 0–95; six CVD causes (IHD, ischemic stroke, intracerebral haemorrhage, hypertensive heart disease, rheumatic heart disease, cardiomyopathy & myocarditis) plus type-2 diabetes; calibrated to GBD 2023; demography and life expectancy from UN WPP 2024. **Data provenance:** every ordinary-run number is read from the current-run results contract (`output/09_deck_results.rds`) and the workbooks' literal probability-of-death series; the cascade strategy is read from its isolated workbook; scenario identity follows the `Scenario_Catalog` contract (binding `include_flag`). No Excel recalculation is used.

Appendix 3 — How costs are calculated

Intervention-based costing adapted from the DCP Cost Model of Watkins et al. (2020), from a health system perspective with annual projections. For intervention i, sex/age group s, year t:

$$C_{i,s,t} = PIN_{i,s,t} \times cov_{i,s,t} \times f_i \times uc_i$$

$$\Delta C_{s,t} = C_{s,t} - C_{0,t}$$

$$PV(\Delta C_s) = \sum_{t=2026}^{2050} \frac{\Delta C_{s,t}}{(1+r)^{t-2025}}$$

$$\text{Cost per death prevented}_s = \frac{PV(\Delta C_s)}{\sum_{t=2026}^{2050}(D_{0,t} - D_{s,t})}$$

PIN = people in need (population $\times$ epidemiological need); cov = coverage (baseline or scenario); f_i = services per person per year; uc_i = Indonesia-adjusted unit cost; subscript 0 = baseline (current course), s = scenario; r = 3%/yr discount rate; price year 2023; currency market US$; perspective health system. Deaths D remain **undiscounted** in the cost-per-death calculation. Downstream cost offsets are **excluded**.

Source: *adapted from Watkins et al. (2020); unit costs, coverage, and people-in-need from the input workbooks and* `Annual_Cost`*; demography from UN WPP 2024.* ***Method:*** *annual costs by intervention $\times$ group $\times$ year, differenced against baseline, discounted to the 2025 base year.* ***Preliminary costing results:*** *unit costs, service use, and delivery assumptions require further validation against BPJS and Ministry of Health data before policy use.*

Appendix 4 — References

GBD 2023 Cardiovascular Diseases and Risks Collaborators. 2025. "Global, Regional, and National Burden of Cardiovascular Diseases and Risk Factors in 204 Countries and Territories, 1990–2023." *Journal of the American College of Cardiology* 86 (22): 2167–2243.

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Watkins, David A., Jinyuan Qi, Yoshito Kawakatsu, Sarah J. Pickersgill, Susan E. Horton, and Dean T. Jamison. 2020. "Resource Requirements for Essential Universal Health Coverage: A Modelling Study Based on Findings from Disease Control Priorities, 3rd Edition." *The Lancet Global Health* 8 (6): e829–39. https://doi.org/10.1016/S2214-109X(20)30121-2.